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KINEXUS

Subsystem Integration of Kinexus with a focus on the design of a Mass Optimised Antenna Monut and Thermal Management Systems

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Abstract

Kinexus presents a novel solution inspired by the NASA DART mission to the oncoming threat of Near Earth Object impact with Earth. To achieve this mission, there are various subsystems within the spacecraft to manage control, power, stability etc. These subsystems are critical to the operation of the satellite. This report outlines the design of structures to support and house the subsystems within the spacecraft taking into consideration the packing volume and individual requirements. Within the report, to demonstrate the design principles that would be applied to design the secondary structures to integrate all of the subsystems on the spacecraft, the design for an antenna mount is presented. The final antenna mount is a titanium cone, with topology-optimised cut outs; it obtains a safety factor of 17.7 in yield and 160.1 in buckling and saves 96% mass as compared to the initial titanium cone. To cool the spacecraft, the report also conceptually highlights the design of 20 m^2 foldable radiators with a Zinc Oxide and Potassium Silicate coating, and to account for temperature variations presents a concept architecture that utilises conduction straps and heat pipes. Ultimately, although touching on multiple varied considerations, the report presents the challenges and potential solutions to the task of Subsystem Integration.

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1 introduction

Over the past two decades [3], knowledge of near-earth objects has increased exponentially. Thus, highlighting the considerable threat posed by a possible asteroid impact [4]. As a consequence, research has been carried out to attempt to deflect incoming asteroids such as NASA's Double Asteroid Redirection Test (DART) [5], wherein a satellite collided with a member of a binary asteroid system using a Kinetic Impactor approach. Project Kinexus presents a proposed satellite constellation that can be deployed to re-direct an incoming, life-threatening asteroid. This report will focus on the integration of the subsystems such as the reaction wheels, sensors and antennae on board the spacecraft.

2 Conceptual Design

The spacecraft consists of 3 modules that have been mounted on top of one another, with a maximum diameter of 2900 mm selected so as to allow 3 spacecraft to fit inside the Starship fairing. The 3 modules are defined as Ariane Grande— which contains the propulsion subsystems; the launch vehicle adapter— which is the structural interface between the lower and upper structures—; and CAN— which houses the avionics of the spacecraft. The exact layout and dimensions are shown in Appendix A.

To analyse these structures, they have been further subdivided into primary, secondary and tertiary structures, wherein each is designed to take a specific load case, hence allowing each structure to be individually optimised. Regarding secondary structures, there are the component mounts, the skin and the deployable structures such as solar panel hinges. Component mounts and their overall layout fall under the scope of this report.

3 Load Cases

Across the flight envelope, the main loads that the component mounts are designed to resist are the inertial loads from the acceleration of the components in the spacecraft [4][6]. Transportation loads have been neglected due to their typically low amplitude [4] and also through the assumption that sufficient damping washers have been included to mitigate the loads. Whereas loads such as random vibrations and space debris have been assumed to be managed entirely by the overall structure and the protective skin respectively[6].

Of the inertial loads on the spacecraft, these are largest during either launch or orbital manoeuvres, wherein thrusters fire. Based on the mission profile, the most constraining would be at launch due to the higher amplitude accelerations sustained from Super Heavy [1], compared to the Aestus engine in the spacecraft itself [7]. The Starship accelerations experienced are shown in Figure 1. There is longitudinal and lateral acceleration. For analyses, the longitudinal loads are said to be equivalent to the mass of the supported components multiplied by the acceleration acting downwards along the axis of symmetry of the spacecraft. Lateral loads are analysed, assuming that they act on mounting holes in the positive and negative of the two orthogonal directions to the axis of symmetry.

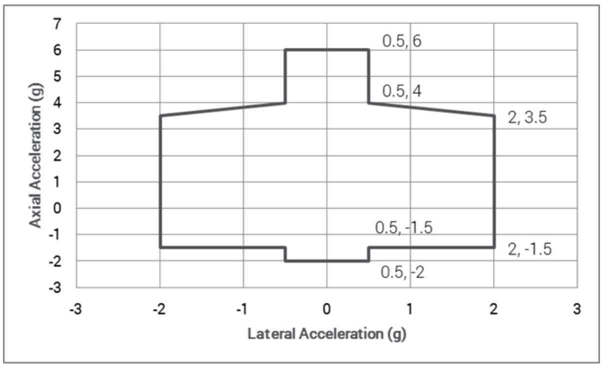

Figure 1: The load envelope for Starship [1]

4 Subsystem Integration

Subsystems are essential to the function of the spacecraft and as a result, their integration and packaging are of critical importance. During integration, compromises must be achieved between conflicting requirements such as: thermal management, volume limitation, and minimising structural mass. Some of the critical subsystem requirements for each subsystem are shown in Table 1, based on [4].

Table 1: A table showing the key subsystems and their requirements

Subsystem	Major Components	Requirements
Power	Batteries, Solar Cells	Cooling and Deployment
Attitude Determination and Control	Sensors, Actuators	Field of View
Telemetry, Tracking and Communication	High Gain Antennae	Gimbal and Stowage
Propulsion	Tanks, Nozzle, Combustion Chamber	Cooling and Fitting
Guidance, Navigation and Control	Fine Guidance Camera	Field of View
Command and Data Handling	Avionics	Cooling

Under the scope of this report, is the integration of subsystems for telemetry tracking and communication (TTC); attitude determination and control (ADC); and ground, navigation and control subsystems (GNC). There is a specific focus on designing an antenna mount, that has been weight-optimised so as to decrease the overall mass of the spacecraft and reach the low structural mass fraction of less than 15% required for the mission[8]. The design methodology used can be further applied to all other mounts in a further piece of work.

4.1 Layout of subsystems

Prior to the design of secondary structures to house the subsystem components, their layout within and around the spacecraft must be defined. This process has been carried out, to maximise the packing volume and attempt to meet the requirements highlighted in Table 1.

Based on analysis of previous spacecraft [9] [6] [5] [10] [11], the subsystems are mounted on thin plate secondary structures. Further, to maintain the temperature of the spacecraft, the high thermal load components like batteries or radio thermal generators are mounted near the outside of the spacecraft [10], allowing irradiation into outer space [12]. Further, heavier components are located near to the centre of mass of the spacecraft. This reduces gravitational gradients which are proportional to the second moments of inertia of the components [13]. Minimising the perturbations from gravity gradients [14] improves the quality of the spacecraft controller.

The power subsystems are the heaviest on the spacecraft and thus are first considered[4]. The batteries weigh 18 kg each [15] and are thus located on the base of the CAN structure. They are placed in a ring near the exterior of the spacecraft to limit their heating effect on high-accuracy sensors within the spacecraft. Solar panels have a considerable impact on spacecraft perturbations, due to their large length, mass and area. Thus are mounted at the base of the Launch Vehicle Adapter, this reduces the moment arm to centre of area and mass. Further, this reduces the need to cut into the Ariane Grande structure, which would greatly reduce the strength of the monocoque [16] [17].

For TTC subsystems, the main consideration is the large high gain antenna, which is required to be Earth Facing[18]. As a result, it must be mounted to the top of the spacecraft. This allows it to transmit signals approximately omnidirectionally [4]. The other consideration is the four low gain antennae, given their small size, they are mounted on opposite sides of the CAN structure, with two per side [18].

In terms of the ADC and GNC subsystems, there are sensors that must be mounted so that they can observe the area around the spacecraft These include the star sensors, star trackers and fine guidance camera. The ideal mounting for the three star trackers, each of them with a 33° field of view, is two in orthogonal directions and a third mounted perpendicular to both [19] To achieve this, two have been fixed to a mounting ring along the centre of the spacecraft. However, for the final star tracker there is the limitation posed by the existence of the nozzle at the base of the spacecraft. Due to the dimensions of the nozzle, the startracker must be mounted at least 501 mm away from the base of the spacecraft. As a consequence, it was mounted on a 1.1 metre-long 2-segment foldable robotic manipulator— to allow it to fold away into the fairing. Due to the presence of the antenna on the top of the spacecraft, a similar, but longer robotic manipulator is used for the fine guidance camera.

On the exterior of the spacecraft, two additional propulsion systems are required. They are reaction control system cold gas thrusters to counteract reaction wheel saturation [20]. And also, small re-phasing thrusters [21]. The cold gas thrusters are attached at a maximum distance from the centre of mass of the spacecraft, thus providing the greatest moment arm. Due to the mounting constraints, the eight are mounted onto two radial ring mounted 3.1 metres above and 2.5 metres below the centre of gravity. Both of the pairs of cold gas thrusters are located 45° offset from the solar panels and radiators to limit any detrimental damage caused by their gas plumes. The re-phasing thrusters are mounted in line with the centre of gravity, to prevent any excessive torques during the manoeuvre, which would consume additional fuel. The tanks themselves are mounted inside the Ariane Grande structure.

The remaining avionics to operate the systems are mounted inside the CAN structure, which has been split into two level to maximise space. On the lower level, the Command and Data Handling subsystems will also be mounted. This greatly limits cabling losses created whilst transmitting power to the high-power draw processors. Further, the reaction wheels are mounted on the lower level, this is due to the pyramid configuration [20] taking up considerable height within the structure. The top level is placed on an annular metal ring. On this ring, the remaining attitude determination equipment is mounted. Cut outs are provided for the star sensors in the walls of CAN. Given the small footprint of the communication avionics, they are mounted on the same plate, this also reduces attenuation losses from transferring the signals to the antenna. This structure is presented in Figure 2.

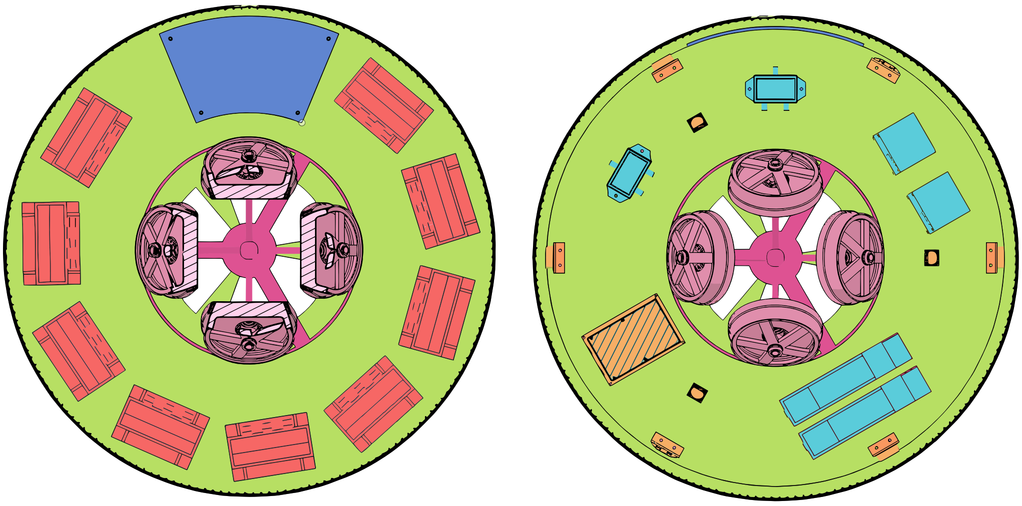

Figure 2: Colour-corrected images showing the layout of the spacecraft internals, showing the bottom level (left) and the top level (right). Blue = Command and Data Handling, Green = CAN Structure, Pink = Reaction Wheels, Red = Power, Orange = ADC and Cyan = TTC

Based on the arrangement of the components, as shown in Figure 3, the centre of gravity (CG), centre of area (CA) and inertia tensor are shown below, based on a coordinate system in metres, with the origin at the base of the nozzle and the x coordinate being upwards, and y and z defined orthogonally according to the right hand rule convention.

$$CG = \begin{pmatrix} 5.8 \\ 0.0 \\ 0.0 \end{pmatrix} \quad (1) \qquad CA = \begin{pmatrix} 9.4 \\ 0.0 \\ 0.0 \end{pmatrix} \quad (2) \qquad Inertial\ Tensor = \begin{pmatrix} 14593 & 217 & 18 \\ 217 & 16807 & 9 \\ 18 & 9 & 25084 \end{pmatrix} \quad (3)$$

4.2 Structural Design Approach

Figure 4 highlights the design methodology that can be followed to design all of the component mounts. This report, will focus on the first iteration of the design of a mass-optimised antenna mount. It is mass optimised to

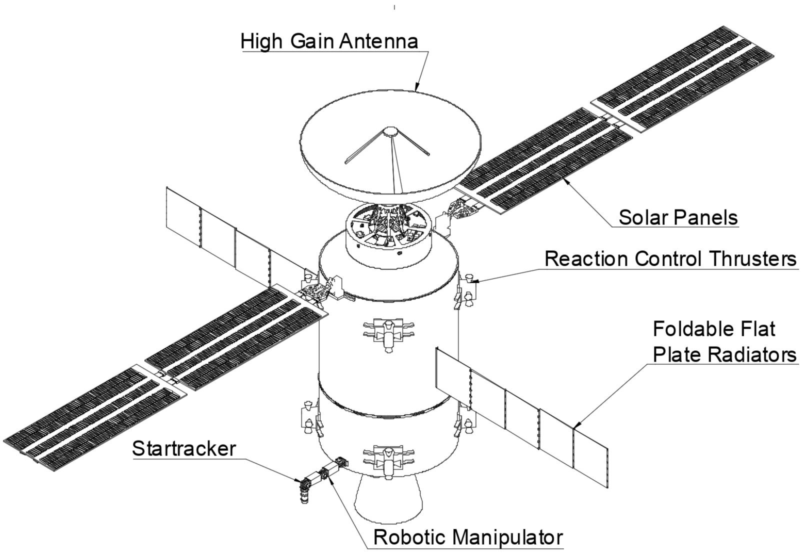

Figure 3: An annotated drawing of the spacecraft showing the placement of key componenrs

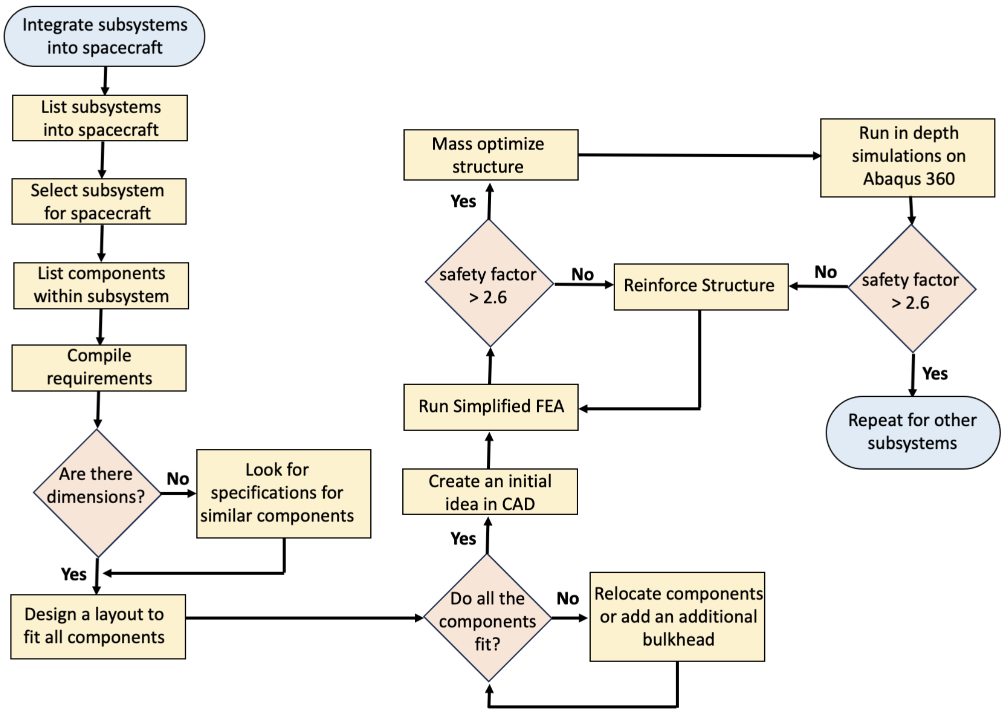

Figure 4: A flow chart demonstrating the design process undertaken

best perform under the critical load cases highlighted in section 3.

4.3 Antenna Mount

Antennae are essential for the spacecraft to communicate with Earth. During the design of the antenna mount, the key constraining factor was the diameter of the fairing (8 m) [1], the necessary antenna gimbal angle (± 70 degrees), diameter (3.77 metres) and height (1.76 metres) [18]. Further, the design of an antenna mount is further restricted because of the considerable inertial mass of the antenna at 20.1 kg.

4.3.1 Initial Design

To design the mount for the antenna dish, similar spacecraft with a large single antenna were studied such as New Horizons [10], Voyager [11], DART [5], etc. Across all of the designs, the single antenna was mounted on a truss structure that was located within the body structure of the spacecraft. However, unlike those spacecraft, there is a much larger gimbal angle required [18]. Further, their mounting approaches would prove unfeasible given the comparatively smaller area present to mount the antenna at the top of the CAN structure.

To successfully gimbal the antenna in orthogonal axes, a ball and socket joint, a pair of revolute joints and the universal joint were considered [22]. Among the three possible options, the pair of revolute joints was chosen due to the ease of manufacture, since it can be made from flat plates, thus reducing the cost of the project, and also allowing parts of high machine tolerance to be manufactured. Additionally, for future work, their analysis is considerably simpler [22]. To actuate the gimbal system, a brushless DC motor [23] was considered to both pitch and rotate the antenna. However, given the mass of the antenna, there is a high torque requirement, which greatly increases mass and/or power draw [24]. Thus, electric linear actuators were considered to pitch the antenna dish which greatly increases the torque that can be provided by modifying the gimbal structure to include protruding arms. Hydraulics were not considered due to the risk of leaking as well as danger of fluid freezing [25] [26]. The low technology readiness level and high torque requirements meant that state of the art actuators like piezoelectrics couldn't be considered [27].

The antenna mount, like all other secondary structures will be manufactured from Ti-6Al-4V which provides a high strength-to-weight ratio, corrosion resistance, among other beneficial properties [2]. The properties are shown in Table 2.

Table 2: A table showing the material properties of Titanium from [2]

Material	Young's Modulus (GPa)	Yield Strength (MPa)	Poisson Ratio
Titanium (6%Al, 4%V)	120	800	0.35

Given the placement of the antenna mount, it necessitates that the structure must be long and slender. Therefore suggesting that buckling and yield must be considered during the launch load case. Based on the construction shown in Figure 5, the antenna mount needs to be greater than or equal to 602.3 mm in length. As a result, a length of 630 mm was deemed appropriate. To allow for the antenna to pivot completely, a maximum diameter of 788.7 mm is required. To limit mass, the diameter was chosen to be 460 mm.

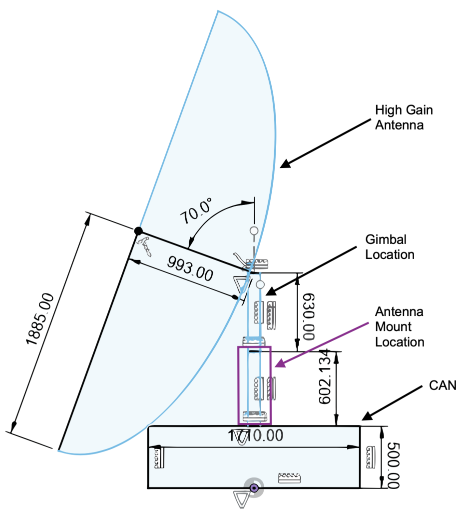

Figure 5: A construction to show the dimensions of the antenna

In terms of the initial structure, a cylindrical shell and a conical shell were considered. This is because either would fully utilise the available area of the structure— and thus maximise the second moment of area, and hence also the rigidity[17]. Further, being axisymmetric means that the structure is resistant to spurious loads that could act in all directions. Based on the structure, wherein the greatest loads are compressive and it is manufactured from an isotropic metal, thus the main failure modes to consider should be buckling and yield. The Euler Buckling load of a 5 mm thick 460 mm diameter cylinder is 2.1 e+09 N [17], Given that the maximum compressive load (from the max longitudinal acceleration case) of the structure is 1183.1 N, this means that the buckling load case is unlikely to be encountered. As a result, during the iterations of the design, the yield load case will be the primary focus.

4.3.2 Cylindrical and Conical comparison

When comparing the structures, the major characteristics are mass, maximum stress and manufacturability. The cylindrical structure is considerably easier to manufacture— it can be manufactured by centrifugal casting, extrusion, rolling etc—, which lowers the cost of the structure, and also reduces the probability of potential defects being introduced that could fail the structure. Whereas the conical structure would certainly be manufactured by rolling a sheet of thin material on a tapered mandrell and welding, thus creating a weakness. [2]

To assess the maximum stress criterion, a 5 mm thick wall cone which tapered from 820 mm diameter to 460 mm was compared with a 5 mm thick wall cylinder of diameter 460 mm. A coarse finite element analysis was conducted in Autodesk Fusion 360— the results are shown in Table 3— to compare the structural behaviour.

Table 3: A table comparing cylindrical and conical designs

Model	Mass (kg)	Max Stress (MPa) for Load Case	
		Max Longitudinal	Max Lateral
Cylindrical	249	29.8	109.1
Conical	377	28.1	24.36

Based on the results above, the conical and cylindrical structures behave similarly in the longitudinal direction, with maximum stress decreasing by only 5.9 % in the conical structure. However, there is substantially better performance in the lateral direction, showing a decrease in maximum stress of 77.7 %. Given that the mass increase of the conical structure is 51.4%, it is, therefore, reasonable to use it as the base design.

4.3.3 Mass Optimisation

In terms of mass optimisation, there are multiple processes that can be utilised such as shape optimisation, size optimisation, topology optimisation and generative design [28] [29]. For the nature of this report, a topology optimisation study and a generative design study will be conducted. These were selected due to the speed of the analyses. Additionally, the process can be modelled in Autodesk Fusion 360, which enables multiple load cases to be modelled simultaneously [30]. This is especially important because as shown in Table 3, structures can show vastly different behaviour in longitudinal and lateral maximum cases. To load the structure, pin constraints were placed on the lower mounting holes, which were modelled as 8 M10 holes. The load was modelled in 2 components: a longitudinal component, modelled as a distributed load acting on a 20 mm diameter circle located around the holes, thus simulating the effect of bolts and washers; and a lateral component modelled as a bearing load that acted in the positive orthogonal directions to the vertical load.

There are three main structural optimisation algorithms: size (where the geometry of the structure is fixed, and the overall dimensions scale); shape (where the outer contours of the shape are varied); and topology (where the density/thickness of the material is altered within a fixed geometry) [28][29]. For this structure, since a cone best utilises the space, topology optimisation has been chosen to create the best lightening holes within the structure. This was done using the Shape Optimisation function in Autodesk Fusion 360[30]. A mass fraction of 30% was set as the target mass fraction, with the objective to maximise stiffness, and thus decrease the stresses on the structure. Within Autodesk Fusion 360, a parabolic mesh using polygons based on a seed size of less than 3% of the model structure was used. The maximum turn angle was defined as 15 degrees. This means that the mesh will be made up of a minimum of 24-sided polygons, wherein each edge is defined using 3 points [31]. Thus allowing it to better fit the curved geometry, thus ensuring that data is not lost during meshing inaccuracies. Two perpendicular symmetry planes were defined along the central axis of the structure so that the structure behaves identically when it is loaded in the 2 directions orthogonal to the central axis. A limitation of similar mass optimisation processes is the fact that the process removes mass not along critical load paths. However, it doesn't calculate the final stresses, and thus final safety factor is unknown.[30]

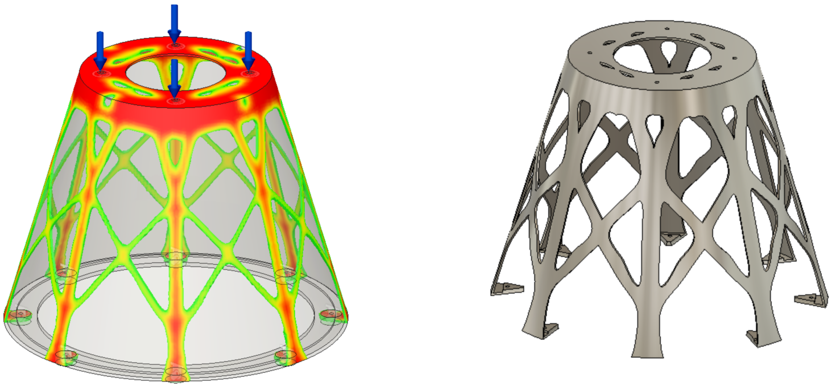

Figure 6: The shape optimisation results (red is highest load, blue is lowest load) (left), and the structure defined based on the results (right)

Generative design is different to the structural optimisation algorithms highlighted above. This is because unlike the above processes which begin with an initially defined structure, Generative Design designs the optimal load-bearing structure based on boundary conditions, preserve geometries, and loads. For the defined structure, the loads and boundary conditions were defined as the inertial loads and the boundary conditions were taken as the pinned mounting holes. A preserve geometry region of 20 mm offset from each of the holes was chosen and a safety factor of 3.0 was defined, this was defined above the typically accepted 2.6 safety factor [32] for untested components, due to the coarse meshing utilised in generative design structures, which may "miss" stress concentrations near holes etc. Symmetry planes have been defined as highlighted above. However, with generatively designed structures, manufacturability is a major concern since the structures are typically designed for manufacture by 3D printing or milling, hence the complex geometries, which cannot be quickly mass-produced, add exceptional cost at the scales required for the spacecraft. [30]

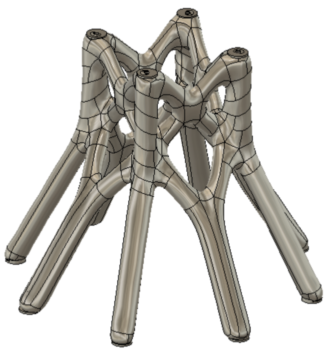

Figure 7: The Generatively Designed Structure

4.3.4 Finite Element Analysis

To compare the structures to one another finite element analyses were conducted. They were conducted initially with Autodesk Fusion 360 using a mesh of quadratic elements [31] with an element seeding size of 2% of the model structure, and a maximum turn angle in the polygons of 15°. This is a relatively fine mesh. However, due to the lack of mesh controls available in Autodesk Fusion 360, a further verification study on the results was conducted using Abaqus. Abaqus enables greater control of physical mesh size [33], and thus allows a more accurate mesh convergence study to be conducted.

Based on the results shown in Figure 8, the loads on both structures are well within the recommended safety margins for spacecraft components as highlighted in [32]. Based on these results, the safety factors are 27 and 250 respectively for the topology optimised and generatively designed structures. As a result, further weight savings could be taken from both structures in second and third iterations of the structure.

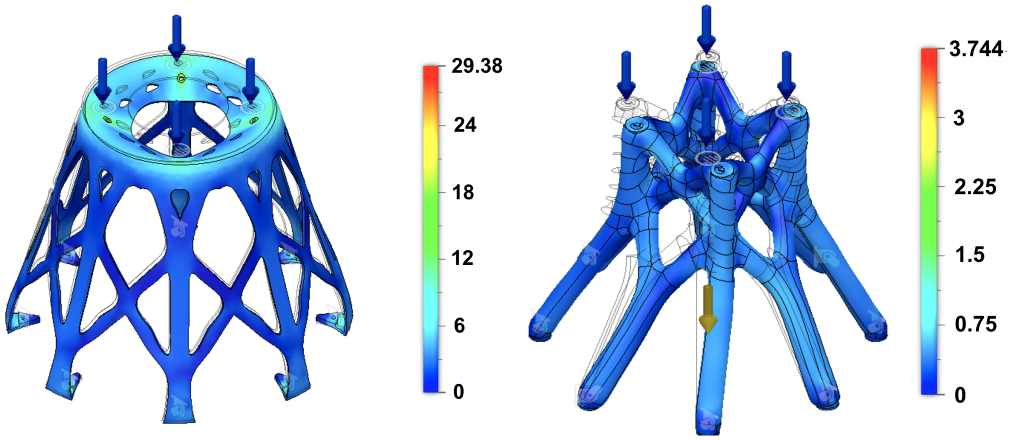

Figure 8: Fusion stress distributions based on the structures (left) Shape Optimisation (right) Generative Design (units MPa)

To verify these results, the loads were inputted into an Abaqus simulation, due to the inability of Abaqus to directly model bearing loads [33], therefore the structure was loaded using a multi-point constraint where the primary node was defined as a central reference point located along the central axis of the structure, and the secondary nodes were defined as the inner area of the holes. Despite this doesn't model the distributed load from the bolt heads, this creates a worst-case scenario, given the stresses will be higher. Further, this enabled a single applied concentrated force on the primary node to model both the lateral and longitudinal accelerations, which simplifies the analysis. The multi-point constraint was defined as a tie constraint, which connects all degrees of freedom and behaviour of the secondary node to that of the primary node[33]. This was used instead of a beam constraint, which would incorrectly, assume a rigid beam connected between the structures [33], thus creating an additional load between the nodes.

In terms of the mesh, a tetrahedral mesh was used. This was chosen due to the complex geometries involved, which could therefore not be meshed by a preferred hex mesh [33], which creates a uniform mesh distribution that lowers the number of elements required to model the structure [31]. This improves solver efficiency. The tetrahedral mesh is a simple solution that can model these complex geometries. A mesh convergence study was carried out for the tetrahedral mesh as shown in Figure 9. Based on the results of the mesh convergence study, as the elements increase above 50000 elements, the results converge. However, there is the added expense of computing power, where the computational time increased by 200% between 60,000 and 100,000 elements, as well as the Education License limit at 250,000 nodes [33]. As a consequence, the study on both structures was conducted using approximately 60,000 elements.

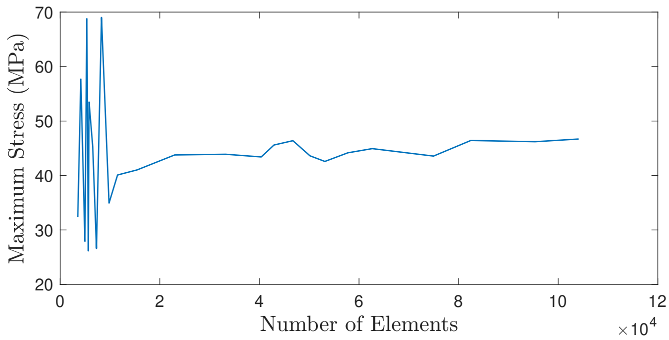

Figure 9: A mesh convergence study based on the shape optimised structure at the maximum longitudinal load case using tetrahedral meshes.

The results of the finite element analyses in Abaqus are shown in Figure 10 and Table 4. From these results, the longitudinal load case is the most constraining, further, in all cases, the structure still behaves with a minimum safety factor of 17.7 and 1250. Both of which, are far above the ideal safety limits for this structure. The maximum stress concentrations are, as expected, around the holes. Additionally, based on the abaqus results, it is clear the meshing discrepancies created by the Autodesk Fusion 360, especially in the case of the generatively designed structure, where unexpected stress concentrations were created, thus lowering the safety factor.

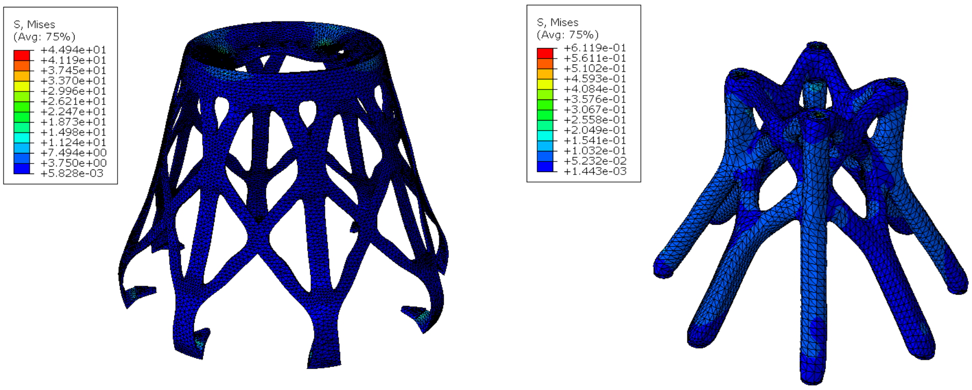

Figure 10: The Abaqus Simulation Results for the Topology Optimised structure (Left) and the Generatively Designed Structure (Right)

Table 4: A table summarising Abaqus Results for all Load Cases

Load Case	Acceleration (g)		Maximum Stress (MPa)	
	Longitudinal	Lateral	Topology Optimised	Generative Design
1	6	0.5	45.12	0.61
2	3.5	2	29.37	0.64
3	-1.5	2	20.31	1.17
4	-2	0.5	15.63	0.45

To provide further validation of the behaviour of the structure, a buckling analysis was also conducted. The buckling analysis was carried out using load case 1. This is because it is the most constraining load case in terms of yield, and further, there are the greatest compressive loads which will cause buckling. The lateral load has been included so that it would allow study of the potential excitation of buckling through bending. In terms of the analysis methods, the Subspace method [27] was chosen over the Lanczos method [34] due to the increased efficiency in solving large eigenvalue problems [33] [35] like the analysis of the buckling of the overall structure. Based on the analyses, an eigenvalue of 160.41 and 47560 was observed for the topology optimised and generatively designed structures respectively. Non-linear buckling analyses have not been considered since the point of the study is to model buckling caused by plastic deformation [17]. However, given the high safety factor in yield and buckling of the structure, one can effectively assume that plastic buckling would occur between the two cases.

Based on the above highlighted analyses, the topology-optimised structure was chosen as the final design, due to the fact that though both structures are heavily over-designed. The topology-optimised structure shows weight savings of 96% compared to 84% with the generatively designed structure.

5 Passive Thermal Management

In a spacecraft, the various subsystems produce considerable heat during operation. This is due to losses caused by resistance along electrical conduction paths. As a result, the components of the spacecraft increase in temperature during operation. Further, there is additional heating from the Sun, reflected solar radiation and infra-red radiation from the Earth, which contribute to heating of the spacecraft [4] [12] [36]. For the designed spacecraft's orbit of 80,000 km [37], the impact of radiation from Earth is negligible and is thus not considered in calculations. Due to the heat balances, mechanisms must be taken to either cool or heat the spacecraft components so that they operate in the appropriate temperature range. For the internal components of the spacecraft a target operating temperature of 10°C has been chosen based on the ideal operational temperatures for all components [4] [38].

5.1 Radiator Sizing

There are many ways to manage the temperature of the spacecraft. Equation 4 and Equation 5 [4] [38] highlight how the heat flux is modelled.

$$Q_{out} = \epsilon\sigma AT^4 \tag{4}$$

$$Q_{in} = \alpha SA_p \tag{5}$$

Where the heat emitted from components can be estimated using Equation 4, which is dependent on emissivity (ϵ), the Stefan Boltzmann constant (σ), external area (A) and temperature difference (T). The heat absorbed can be calculated by Equation 5, where α is the absorptivity of the spacecraft, S is the solar constant and A_p is the projected area.

In terms of thermal management, the major contributors are the coatings on the exterior of the spacecraft like MLI or plasma electrolytic coatings and the radiator sizing. Based on estimates for the thermal loads required on the spacecraft as illustrated in [37], the spacecraft requires both heating and cooling. To maintain the temperature of components at 5 au [37], 700 W of heating are required, and thus a heater was chosen. However, at 1 au, for the parking orbit of the spacecraft, 2000 W of heat must be irradiated from the spacecraft.

For the radiator of the spacecraft, the flat plate geometry has been chosen due to its simplicity in manufacture, large surface area and also foldability. [12] [36]. Despite, the benefits, there are issues in the fact that it increases the length of the radiator, thus impacting the stability requirements of the spacecraft, due to the increased distance from the centre of mass [13]. The radiators are foldable to enable active thermal control and stow away during launch. This has been chosen over louvres [12] to maximise heat dissipation area and also to prevent creating cut-outs in the exterior of the Ariane grande structure. Further, given the slats in a louvre, a failure of one of the motors would prove destructive to the spacecraft system. The radiators would be mounted at the centre of area of the spacecraft without radiators so that as they alter their geometry by folding in and out, they do not affect solar radiation perturbations on the stability of the spacecraft.

When considering the coating of the radiators, the major considerations, especially for this project, are degradation [39] [40], emissivity and absorptivity. The degradation of the coating on the radiator occur through UV radiation and interactions with energised particles; through outgassing of the spacecraft materials, that absorbs contaminants into the coating and; also, from thermal cycling loads causing damage [41]. These can considerably decrease the emissivity and increase absorptivity of materials [39]. Materials can show increases of up to 40% under high exposure to UV radiation [42]. This is particularly important because the radiators should perform nominally over its entire lifetime (15 years)[37]. As a consequence, a Zinc Oxide and Potassium Silicate coating has been chosen, this is due to its constant performance over long hours of exposure [41] [42]. Based on an emissivity value of 0.17 and absorptivity of 0.92 [4] [43], a radiator area of 9.1 m^2 is required at 1 au. However, to allow the spacecraft to approach nearer to the sun, this has been doubled to 20 m^2. Each radiator will be made from 5 1 metre by 1 metre panels, this allows them to fold and stow around the spacecraft, and thus fit inside the fairing.

5.2 Thermal Management of Components

Once the components have been placed inside the spacecraft, analyses can be carried out to determine how to manage the different temperature requirements of various different components. From radiation and conduction, the ambient temperature in the spacecraft varies considerably [12] [40]. Further, the temperature needs of typical components varies from minimums of -100ºC to 0ºC, and maximums of 15 to 100ºC [4]. For highly sensitive components like batteries whose operating temperature is between 0 and 15ºC, temperature must be carefully monitored [36].

For this spacecraft, given the distribution of area, this means that each section receives different amounts of solar radiation. This distribution is shown in Table 5.

As shown in Table 5, the majority of heat is absorbed in the solar panels, followed by the Ariane Grande Structure. This is due to their considerable size. The solar panels also have an abnormally high absorptivity [44] given their

Table 5: A table showing the heat absorbed and rejected by each component at 1 au

Section	$Q_{in}(W)$	$Q_{out}(W)$
CAN	198.3	41.6
LVA	385.0	1250.8
Ariane Grande Upper	2891.0	225.6
Ariane Grande Nozzle	1297.1	336.6
Radiator	1535.3	6808.1
Solar Panels	51402.1	7329.5
Antenna	5544.1	2210.8

function of absorbing solar radiation. As a result, mechanisms must be included to better distribute the heat throughout the spacecraft. This can be achieved through the usage of conductance straps, heat pipes, etc [12] [36][40] . For this spacecraft, given their reliability, due to the lack of mechanical components, heat pipes have been chosen. Heat pipes transfer thermal energy using capillary action [4]. However, the main issue with heat pipes is their cost, and usage of space [40]. These can be implemented in this spacecraft given the fact that the system can be mounted to the exterior of the spacecraft. Heat pipes are essential to transfer temperature to the CAN structure, given the fact that under the current heating of the CAN structure as shown from the Autodesk Fusion 360 thermal simulation in Figure 11, the highest temperature (red) is -42.9°C and the lowest temperature (blue) is -102.9°C, hence the structure is at a lower temperature than the minimum operating temperature of the components. Furthermore, based on the analysis, the thermal gradients can be quite steep up to 97 K/m, this should be analysed in further work since it can cause significant thermal stresses in the structure, which can lead to failure. Additionally, these can contribute to high thermal cycling loads as the spacecraft rotates.

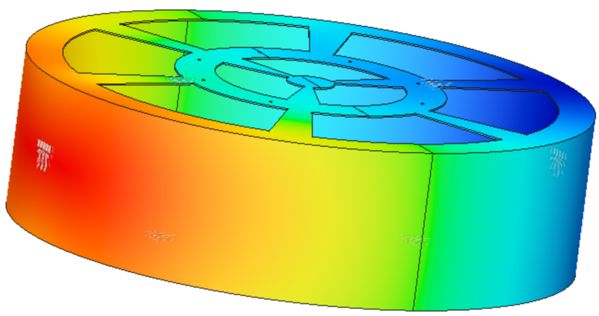

Figure 11: The thermal distribution of CAN, red is maximum temperature, and blue is minimum temperature

To distribute heat within CAN, a network of conducting straps will be located around the heaters and connected to the heat pipes to distribute the thermal heat to the most sensitive components, with a particular focus on the batteries. To manage the temperature an array of thermostats will be mounted on each of the components so that their temperature can be monitored [36], thus determining when to operate the heater as well as fold/unfold the radiators. Further, for the spacecraft, to limit heat losses, an insulative layer will be designed to surround the electronics [12]. This prevents the loss of excess heat through radiation. This is particularly critical for the Star Trackers, which are located outside, operate under very sensitive thermal requirements and are essential for attitude determination. For the insulation a material with low absorptivity and emissivity like Multi-Layer-Insulation [40] must be employed given the fact that the star trackers, will rotate into the direction of the Sun, and thus experience considerable heating as well. The same structure would be used for the fine guidance camera.

6 Discussion and Further Work

This report has presented a “first-pass” of a design to integrate the many subsystems included on the spacecraft. There has been a consideration of the locational and thermal restrictions present within the design, with a primary focus on the structural design of a single component i.e. the antenna mount, which factors in inertial accelerations as well as unique requirements such as gimbal angle.

6.1 Layout

The layout of the spacecraft includes components mounted on the exterior and interior of the spacecraft. The cold gas thrusters and two star trackers are mounted directly to the outside of the Ariane Grande structure, and a third star tracker is mounted on a robotic manipulator at the bottom of the spacecraft. Around the CAN structure, the antenna is mounted on top, and a fine guidance camera is similarly mounted with a robotic manipulator. The CAN structure has been designated as the main avionics mount for the spacecraft, and is divided into two levels, with power subsystems, reaction wheels and CDH subsystems on the bottom level; and on the top level, the main stability sensors are located and also the TTC subsystems.

However, with regard to the layout of the spacecraft, the design presented is limited given the fact that the initial dimensions of the CAN structure were defined prior to the selection of the interior components. As a result, there is considerable wasted space. In a second iteration of the design, the height of CAN could be decreased, to potentially create a more efficient packing structure. Alternatively, the diameter was chosen to fit with the pre-existing launch vehicle adapter, and thus could've been similarly varied. In future, an analysis should be conducted based on the dimensions of the components and determine the ideal packing configuration. This would be similar to solving a variation on the "Knapsack problem".

6.2 Antenna Mount

The antenna mount has been selected using shape optimisation, and it has been designed from a titanium cone. The structure has been simulated in a mesh converged Abaqus study and obtained a safety factor of 17.7 in yield and 160.1 in buckling.

For the design of the antenna mount, given the high safety factors that it currently operates within, there is room for further mass optimisation, and as a result, further investigations could be conducted to observe the effects of changing the thickness of the cone, as well as taking further steps towards mass optimisation, by lowering the desired mass percentage to less than 30%. Furthermore, with regards to the current structure, further simulations could be conducted on it, to verify the structure, these include non-linear buckling analyses, which were neglected due to the pre-existing high safety factors for linear buckling analyses.

Additionally, there are further secondary structures that remain undesigned, this includes designs for the reaction wheel mount, which would need to consider the high torques produced, and also of the reaction control system cold gas thrusters mounted on the outside of the spacecraft. Furthermore, there is potential for considerable mass optimisation to be conducted on the design of the robotic manipulators.

6.3 Passive Thermal Management

For passive thermal management of the spacecraft, on top of the Multi Layer insulation and various coatings applied, there are deployable radiators with a cumulative maximum area of 20 m^2, thus, allowing a maximum of 4.3 kW to be dissipated at one time. Further, on the spacecraft, the avionics will be located within separate insulated casings, so as to thermally isolate them from the large temperature variations observed in the exterior of the spacecraft.

In addition, there is considerable potential to expand upon the conceptual discussion of thermal systems presented within this report. With the radiator design, there is potential to experiment with different cross sections and profiles instead of the standard flat plate radiator presented. Further, analyses could be conducted to design weight-optimised hinges, which are capable of taking the mass of the radiator. Additionally, analyses could be conducted to explore the stresses induced through thermal cyclic loads, as well as better characteristics of the thermal distributions in the structure. Furthermore, with regard to the internal system, there is room to size the heat pipes, conductance straps and insulative housings directly got this design, based on the heat fluxes present on the walls, so that all of the components operate in ideal conditions.

Despite, there still remaining room for further work, this report outlines a sufficient conceptual design of subsystem integration, and contains frameworks, that can be used as the basis for further designs.

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Appendix

7 Appendix A

The configuration of the spacecraft is shown below. In it, the breakdown and dimensions of the individual modules can be seen.

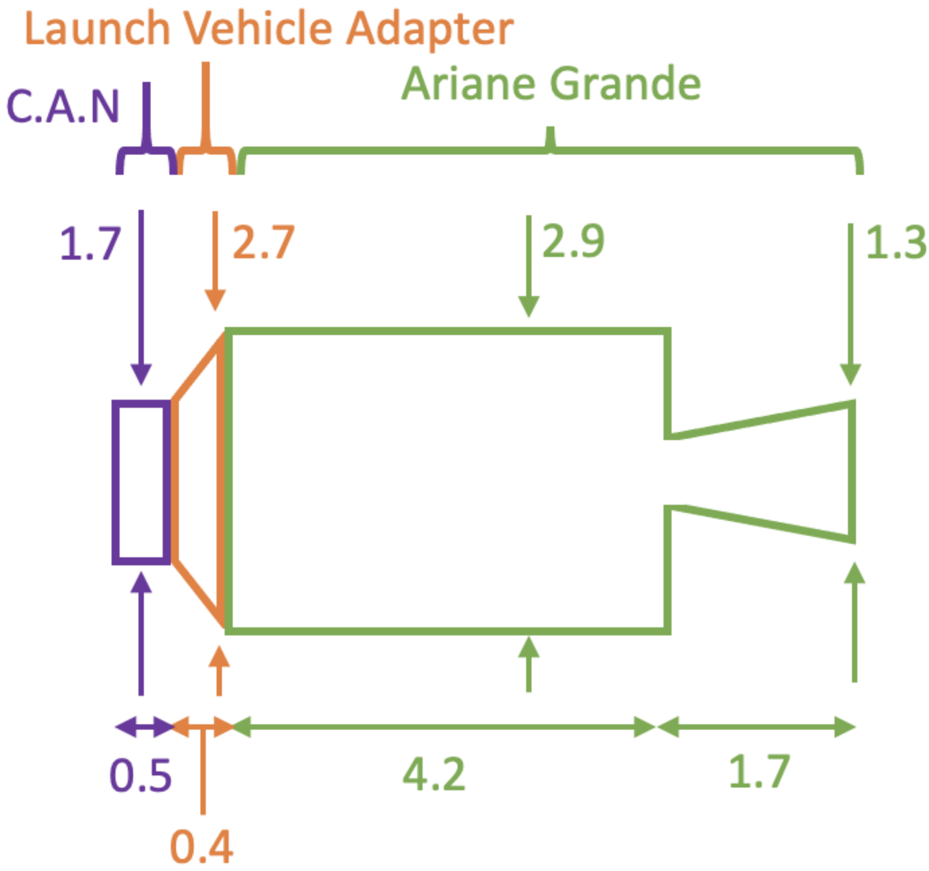

Figure 12: The structure of the spacecraft, highlighting the location of the 3 modules